Date :

Functions

Let 'A' and 'B' be non-empty sets. A relation from 'A' to 'B' such that to every element 'a' of 'A' there corresponds a unique element 'b' of 'B', is called a function f from A to B.

A function from A to B assigns each element of A to a unique element of B. It is denoted as $f: A \to B$

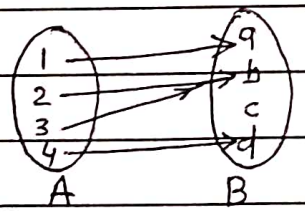
If the element a of A under f corresponds to the element b of B, we shall write $f(a) = b$.

The element 'a' is called the argument or pre-image and 'b' is called its value or the image of 'a' under f.

The set of elements of A is called the domain. The set of elements of B which are images or values of the elements of A is called the range of the function. The set B is called the codomain. The range is a proper subset of or equal to codomain.

Let $A = \{1,2,3,4\}$; $B = \{a,b,c,d\}$ and let $f = \{(1,a),(2,b),(3,b),(4,d)\}$

Here we have $f(1) = a$, $f(2) = b$, $f(3) = b$, $f(4) = d$

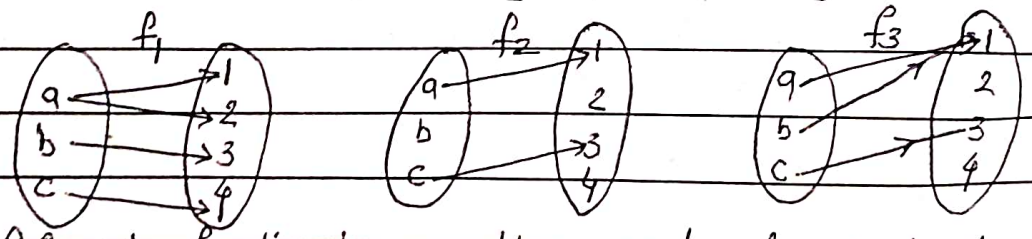


$Dom(f) = \{1,2,3,4\}$

$Codom(f) = \{a,b,c,d\}$

$Ran(f) = \{a,b,d\}$

Example:- let $A = \{a,b,c\}$, $B = \{1,2,3,4\}$ and let $f_1 = \{(a,1),(a,2),(b,3),(c,4)\}$ and $f_2 = \{(a,1), \cancel{(b,1)}, (c,3)\}$, $f_3 = \{(a,1),(b,1),(c,3)\}$



f_1 is not a function because there are two elements 1 and 2 in B which corresponds to the same element 'a' of A.

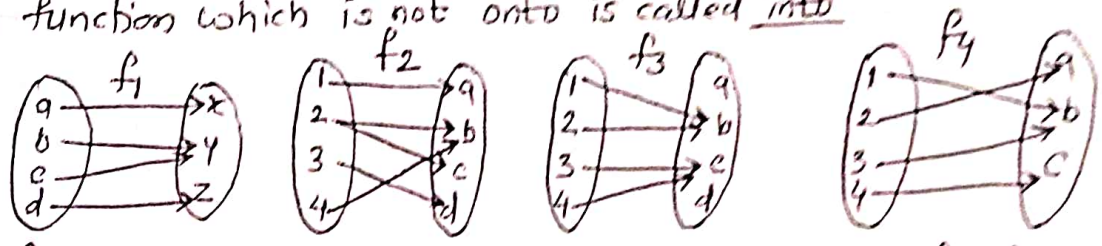
f_2 is also not a function because the element b of A is not associated with any element of B

f_3 is a function from A to B.

$fofof = f(f(f(x))) = f(f(x+2) = f(x+2+2) = f(x+4) = x+6$

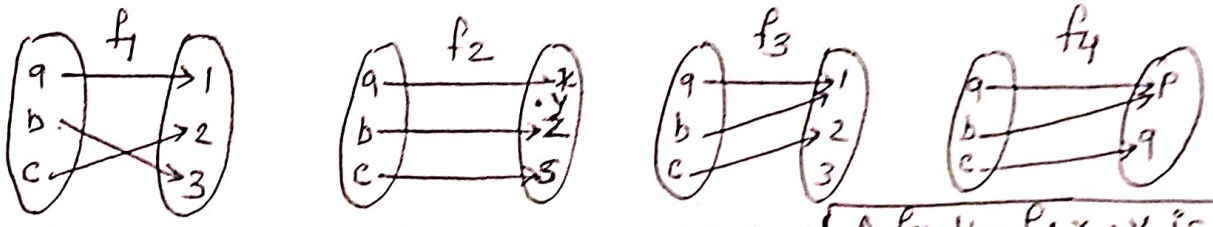
Types of functions:-

→ A function $f: A \to B$ is called <u>surjective</u> or <u>onto</u> if every element of $b \in B$ is an image of at least one element a of A.
In other words, the range of $f = B$.
A function which is not onto is called <u>into</u>



f_1 is onto or surjective because every element x, y, z of B is an image of at least one element of A
f_2 is not a function
f_3 is not onto because there is no arrow to a or to d
f_4 is onto or surjective

→ A function f is said to be <u>injective</u> or <u>one-to-one</u> if two distinct elements of A corresponds to two distinct elements of B.
i.e. each element of set A is mapped with a unique element of B.



f_1 is one-to-one and onto (called bijective)
f_2 is one to one but not onto
f_3 is not one-to-one and not onto
f_4 is onto but not one-to-one

> A function $f: x \to y$ is one-to-one or injective if $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$ or $f(x_1) = f(x_2)$ implies $x_1 = x_2$

→ If $f: A \to B$ is <u>both</u> <u>one-to-one</u> <u>and</u> <u>onto</u> then f is called <u>bijective</u>

Example:- comment whether the function f is one-to one or onto. consider function: $f: N \to N$ where N is set of Natural numbers including zero. $f(j) = j^2 + 2$

Sol^n:- $N_1 = \{0,1,2,3,4 \ldots\}$ & $N_2 = \{0,1,2,3,4 \ldots\}$ $\therefore f: N_1 \to N_2$

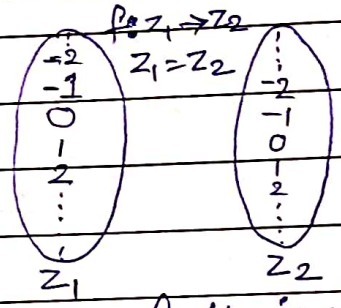
$f(0) = (0)^2 + 2 = 2$
$f(1) = (1)^2 + 2 = 3$
$f(2) = (2^2) + 2 = 6$
$f(3) = (3)^2 + 2 = 11$
$f(4) = (4)^2 + 2 = 18$
.... and so on

for every 'n' from N_1, we can find another 'n' of set N_2. So the given function is one-to-one.
But not every element 'n' of set N_2 is image of elements from set 'N_1'. So the given function is not onto.

Example: Test whether the following function is one-to-one, onto or both.

$f: Z \to Z,\ f(x) = x^2 + x + 1$

Sol^n:- A function from A to B is one-to-one, if no two elements of A ~~should~~ have same image.



~~let~~ $f(x) = x^2 + x + 1$

$f(-2) = (-2)^2 - 2 + 1 = 3$

$f(-1) = (-1)^2 - 1 + 1 = 1$

$f(0) = (0)^2 + 0 + 1 = 1$

$f(1) = 1^2 + 1 + 1 = 3$

$f(2) = 2^2 + 2 + 1 = 7$

$f(3) = 3^2 + 3 + 1 = 13$

So elements 1 & -2, 0 & -1 have same image so the function is not one-to-one.

The given function is not 'onto' because every element of Z_2 is not image. (2, 4, 5, 6 are not images)

So the above function is not one-to-one nor onto. therefore not bijective.

Example – Let $f: R \to R$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$ find $\{f o f^{-1}\}(x)$.

Sol^n (MAY 19 & DEC 19) →

$$\{f o f^{-1}\}(x) = f(f^{-1}(x)) = f\left(\frac{x+1}{2}\right)$$

but $f(x) = 2x - 1$

$$\therefore f\left(\frac{x+1}{2}\right) = \cancel{2}\left(\frac{x+1}{\cancel{2}}\right) - 1$$

$$= x$$

(MAY 22)

Example:- Show that mapping $f: R \to R$ given by $f(x) = 4x - 3$ & $f(x) = 4x + 7$ is bijective

Sol^n:- ① $f(x) = 4x - 3$

for one-to-one

lets $f(x_1) = f(x_2)$

$4x_1 - \cancel{3} = 4x_2 - \cancel{3}$

$\cancel{4}x_1 = \cancel{4}x_2$

$\therefore x_1 = x_2$

then function is one-to-one

for onto, $f(x) = y$

$\therefore y = f(x) = 4x - 3$

$y = 4x - 3$

$x = \frac{y+3}{4}$

$\forall y \in B, \exists x \in A$

for every value of y of B, there exists x from A

therefore the function is onto

Since the function is one-to-one & onto

$\therefore$ The function is Bijective

Since the function is bijective inverse exist

$y = f(x)$

$\therefore x = f^{-1}(y)$

$$\boxed{x = \frac{y+3}{4}}$$

Same procedure for $f(x) = 4x + 7$

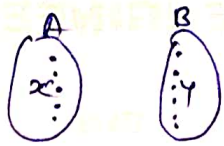
$fofof = f(f(f(x))) = f(f(x+2)) = f(x+2+2) = f(x+4) = x+6$

{α, β, , C, γ

Graphica

Example:- Let $f: R-(7/5) \to R-(\frac{2}{5})$ be defined by $f(x)=\frac{2x-3}{5x-7}$. Whether the function is bijective. Justify your answer.

Example:-



'x' should not have value $(\frac{7}{5})$ because at '7/5' f(x) becomes '∞'

Same way y should not have value '2/5'

for one-to-one, let $f(x_1) = f(x_2)$

$$\frac{2x_1-3}{5x_1-7} = \frac{2x_2-3}{5x_2-7}$$

Cross multiply

$$\therefore 10x_1x_2 - 14x_1 - 15x_2 + 21 = 10x_1x_2 - 15x_1 - 14x_2 + 21$$

$$10x_1x_2 + 15x_1 = 10x_1x_2 + 15x_2$$

$$15x_1 = 15x_2$$

$$\therefore x_1 = x_2$$

function is one-to-one

$f(x) = y$

$\therefore y = f(x)$

$y = \frac{2x-3}{5x-7}$

$\therefore 5xy - 7y = 2x - 3$

$5xy - 2x = 7y - 3$

$x(5y-2) = 7y-3$

$\therefore x = \frac{7y-3}{5y-2}$

for every value of 'y' in B, x must exists.

But for $y = \frac{2}{5}$, x does not exist

$\forall y \in R-(\frac{2}{5}); \exists x \in R-(\frac{7}{5})$

$\therefore$ function is onto

Since function is one-to-one and onto, function is bijective.

As function is bijective then f^{-1} exist

$y = f(x)$

$$\therefore x = f^{-1}(y) = \frac{7y-3}{5y-2}$$

Example:- Let $A = B = R$, the set of real numbers. Let $f: A \to B$ given by $f(x) = 2x^3 - 1$ and let $g: B \to A$ given by $g(y) = \sqrt[3]{\frac{1}{2}y + \frac{1}{2}}$ show that f is bijection betn A to B & g is bijection betn B to A

→ $f: A \to B$, check for one-to-one

$f(x_1) = f(x_2)$

$\therefore 2x_1^3 - 1 = 2x_2^3 - 1$

$\therefore \boxed{x_1 = x_2}$ f^n is one to one

Now. $f(x) = y$ $\therefore y = f(x)$

$y = 2x^3 - 1$ $\therefore 2x^3 = y + 1$

$\therefore x^3 = \frac{y}{2} + \frac{1}{2}$ $\therefore x = \sqrt[3]{\frac{y}{2} + \frac{1}{2}}$

$\therefore \forall y \in B; \exists x \in A$

f^n is onto

$\therefore$ f is bijective function betn A and B

Similary for $g: B \to A$,

check for one to one

$g(y_1) = g(y_2)$

$\therefore \sqrt[3]{\frac{1}{2}y_1 + \frac{1}{2}} = \sqrt[3]{\frac{1}{2}y_2 + \frac{1}{2}}$

$\frac{1}{2}y_1 + \frac{1}{2} = \frac{1}{2}y_2 + \frac{1}{2}$

$\therefore \boxed{y_1 = y_2}$ g is one-to-one

Now $g(y) = x$

$x = g(y) = \sqrt[3]{\frac{1}{2}y + \frac{1}{2}}$

$x^3 = \frac{1}{2}y + \frac{1}{2}$ $\therefore 2x^3 = y + 1$

$\boxed{\therefore y = 2x^3 - 1}$

for each 'x' in A, there is corresponding y in B

$\forall x \in A; \exists y \in B$

$\therefore$ g(y) is onto function

Since g is one-to-one & onto, g is bijective f^n betn B & A

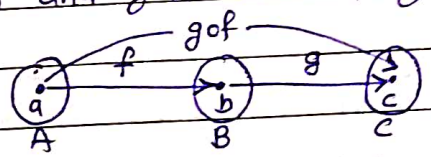
~~Since~~

Compositions

Let f be the function of A to B (ie. $f: A \to B$) and g be the function from B to C (ie $g: B \to C$). Then the composition of f and g denoted ~~by~~ as gof is a relation from A to C, where $gof(a) = g(f(a))$

$gof: A \to C$ is a function

If $f(a) = b$ and $g(b) = c$ then $gof(a) = c$



Ex:- Consider $f(x) = 2x - 3$ find formula for a composition function.

(i) $f^2 = fof$ and (ii) $f^3 = fofof$

Soln:- (i) $(fof)(x) = f(f(x)) = f(\underbrace{2x-3}_{x}) = 2(2x-3)-3 = 4x-9$

(ii) $(fofof)(x) = f(f(f(x))) = f(f(\underbrace{2x-3}_{x})) = f(2(2x-3)-3) = f(\underbrace{4x-9}_{x})$

$= \cancel{4(4x-3)-9} = 2(4x-9)-3 = 8x-21$

Ex:- Let $f: R \to R$, $f(x) = x^2 - 1$, $g(x) = 4x^2 + 2$ find $fo(gof)$ & $go(fog)$

Soln:- $fo(gof) = fo(g(f(x))) = f(g(\underbrace{x^2-1}_{x})) = f(4(x^2-1)^2+2)$

$= f(4(x^4-2x^2+1)+2) = f(4x^4-8x^2+6)$

$= (4x^4-8x^2+6)^2 - 1$

$go(fog) = g(f(g(x))) = g(f(\underbrace{4x^2+2}_{x})) = \cancel{g((4x^2+2)^2-1)}$

$= g((4x^2+2)^2-1) = g(16x^4+16x^2+4-1) = g(\underbrace{16x^4+16x^2+3}_{x})$

$= 4(16x^4+16x^2+3)^2+2$

Ex:- If $f: R \to R$ and ~~g(x)~~ $g: R \to R$ defined by $f(x) = x+2$ & $g(x) = x^2$ find fgf & gfg

Soln:- $fogof = f(g(f(x))) = f(g(\underbrace{x+2}_{x})) = f(x+2)^2 = (x+2)^2+2 = x^2+4x+6$

$gofog = g(f(g(x))) = g(f(x^2)) = g(x^2+2) = (x^2+2)^2 = x^4+4x^2+4$

Ex:- Let $f: R \to R$ defined as $f(x) = x^3$ & ~~g(x)~~ $g: R \to R$ defined as $g(x) = 4x^2+1$ find out gof, fog, f^2, g^2

Soln:- $gof = g(f(x)) = g(x^3) = 4(x^3)^2+1 = 4x^6+1$

$fog = f(g(x)) = f(\underbrace{4x^2+1}_{x}) = (4x^2+1)^3$

$fof = f(f(x)) = f(x^3) = (x^3)^3 = x^9$

$gog = g(g(x)) = g(4x^2+1) = 4(4x^2+1)^2+1 = \cancel{8x^2+}$

Ex:- Let $f(x) = x+2$, $g(x) = x-2$ & $h(x) = 3x$ for all $x \in R$ find

$fogoh = f(g(h(x))) = f(g(3x)) = f(3x-2) = 3x-2+2 = 3x$

$hogof = h(g(f(x))) = h(g(x+2)) = h(x+2-2) = h(x) = 3x$

$fofof = f(f(f(x))) = f(f(x+2)) = f(x+2+2) = f(x+4) = x+6$

Ex:- Let $f(x) = 2x+3$, $g(x) = 3x+4$, $h(x) = 4x$ for all $x \in R$

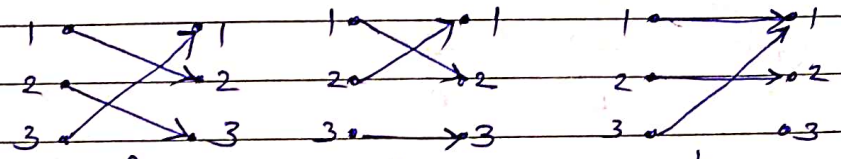
find gof, fog, foh, hof, goh [Ans:- $6x+13$, $6x+11$, $8x+3$, $8x+12$, $12x+4$]

Example:- functions f, g, h are defined on a set $x = \{1,2,3\}$ as $f = \{(1,2)(2,1),(3,1)\}$

$g = \{(1,2),(2,1),(3,3)\}$, $h = \{(1,1),(2,2),(3,1)\}$

find, fog & gof are equal. find $fogoh$ and $fohog$

Solution:-



f g h

$f(1) = 2$ $g(1) = 2$ $h(1) = 1$

$f(2) = 3$ $g(2) = 1$ $h(2) = 2$

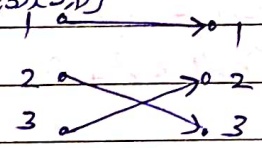
$f(3) = 1$ $g(3) = 3$ $h(3) = 1$

$gof(1) = gof(x) = g(f(x)) \Rightarrow g(f(1)) = g(2) = 1$

$gof(2) = g(f(2)) = g(3) = 3$

$gof(3) = g(f(3)) = g(1) = 2$

$gof = \{(1,1),(2,3)(3,1)\}$

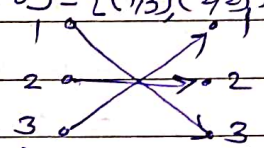


$fog(x) = fog(1) = f(g(1)) = f(2) = 3$

$fog(2) = f(g(2)) = f(1) = 2$

$fog(3) = f(g(3)) = f(3) = 1$

$fog = \{(1,3),(2,2),(3,1)\}$



$fog \neq gof$

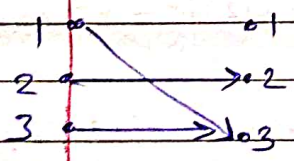
$fogoh(x) = f(g(h(x)))$

$fogoh(1) = f(g(h(1))) = f(g(1)) = f(2) = 3$

$fogoh(2) = f(g(h(2))) = f(g(2)) = f(1) = 2$

$fogoh(3) = f(g(h(3))) = f(g(1)) = f(2) = 3$

$fogoh = \{(1,3),(2,2)(3,3)\}$

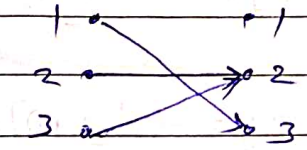


$fohog(1) = 3$

$fohog(2) = 2$

$fohog(3) = 2$

$fohog = \{(1,3),(2,2),(3,2)\}$



Identity:- (I_A)

We always define function as,

$f: A \to A$ (i.e. $B = A$) as $f(a) = a$ for all $a \in A$

then f is called the Identity or Identify function of A and is denoted by I_A $\quad \therefore I_A = \{(a,a) \mid a \in A\}$

Example - Let $A = \{1,2,3\}$ and $f: A \to A$ is identify function.

since $f(1) = 1$, $f(2) = 2$, $f(3) = 3$

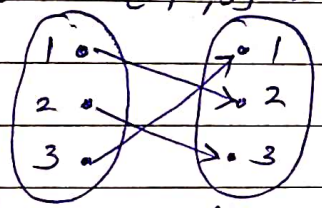
Inverse function:- (F^{-1})

Let $f: A \to B$ be function then

$f^{-1}: B \to A$ is called the inverse mapping of f

If $f: A \to B$ is a bijection and $f(a) = b$, then $a = f^{-1}(b)$

Let $A = \{1,2,3\}$ & 'f' be the func^n on A such that $f(1) = 2, f(2) = 3, f(3) = 1$



$f: A \longrightarrow A$

$f = \{(1,2),(2,3),(3,1)\}$

Then $F^{-1}: A \to A$

$$\left.\begin{array}{l} f^{-1}(1) = 3 \\ f^{-1}(2) = 1 \\ f^{-1}(3) = 2 \end{array}\right\} f^{-1} = \{(1,3),(2,1),(3,2)\}$$

[A function f for which f^{-1} exists called invertible]

Example:- If $f: A \to B$ & $g: B \to C$ are both one-to-one & onto, then $(gof)^{-1} = f^{-1} o g^{-1}$

Sol^n:- If $f: A \to B$ & $f(a) = b$ then $a = f^{-1}(b)$

If $g: B \to C$ & $f(b) = c$ then $b = g^{-1}(c)$

Then $gof(a) = g(f(a)) = g(b) = c$

$\therefore a = gof^{-1}(c)$ — (I)

$\Rightarrow f^{-1} o g^{-1}(c) = f^{-1}(g^{-1}(c)) = f^{-1}(b) = a$ — (II)

combining (I) & (II)

$(gof)^{-1} = f^{-1} o g^{-1}$

Example:- function $f(x) = \dfrac{4x+3}{5x-2}$. Find f^{-1}

Sol^n:- $f(x) = \dfrac{4x+3}{5x-2}$

exchange x and y to get $x = \dfrac{4y+3}{5y-2}$, solve for 'y'

$\therefore 5xy - 2x = 4y + 3$

$5xy - 4y = 2x + 3$

$(5x-4)y = 2x+3 \quad \therefore y = \dfrac{2x+3}{5x-4}$

$\therefore f^{-1} = \dfrac{(2x+3)}{(5x-4)}$